Ahmed Bennasser

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LinkedIn • GitHub • Portfolio

Final-year Engineering Student at Ensimag & MSc in Applied Mathematics (MSIAM), specializing in Machine Learning, NLP, and Generative AI. Proven experience in designing production-ready Agentic RAG pipelines, fine-tuning large models (PEFT/LoRA), implementing hybrid search (BM25 + vector embeddings), and building objective LLM-as-a-Judge evaluation systems. Seeking a 5- to 6-month End-of-Study Internship in LLM / RAG / AI Research Engineering starting February 2027.

EDUCATION

2024 – Present — **Grenoble INP – Ensimag & Université Grenoble Alpes**

- **Master 2 MSIAM** – MSc in Industrial and Applied Mathematics (Data Science & Mathematical Modeling track).
- **Diplôme d'Ingénieur (State Engineering Degree)** – Mathematical Modeling, Image, and Simulation track.

2022 – 2024 — **Lycée Mohammed VI d'Excellence**

- **Classes Préparatoires aux Grandes Écoles (CPGE)** – Intensive curriculum in Mathematics and Physics (MP).

RESEARCH & ENGINEERING EXPERIENCE

Feb 2026 – May 2026 — **Research Intern – LIG Laboratory (GETALP Team), Grenoble**

- Evaluated French multimodal foundation encoders (*Pantagruel*) on downstream NLP and Named Entity Recognition (NER) benchmarks.
- Engineered an end-to-end multimodal pipeline: curated paired datasets, applied parameter-efficient fine-tuning via **PEFT/LoRA**, and used quantization for low-latency, memory-efficient inference.
- Designed an automated hybrid evaluation framework combining mathematical distribution metrics (style, contrast) with an **LLM-as-a-Judge** semantic faithfulness pipeline.
- **Technologies:** PyTorch, Hugging Face Transformers, LoRA, PEFT, Quantization, Stable Diffusion, LLM-as-a-Judge. GitHub

Jun 2026 – Jul 2026 — **Assistant Engineer Intern – LIG / TIMC Laboratories, Grenoble**

- Implemented the *Hierarchical Feature Engineering* method to leverage taxonomic trees on sparse count data.
- Formulated a statistical feature selection algorithm (**SFE**) optimized for high-dimensional regimes ($p \gg n$), validated across 5 protocols (noise stability, inter/intra-study consistency).
- **Technologies:** Python, Scikit-learn, NumPy, Pandas, Inferential Statistics, Optimization.

LLM, AGENTIC RAG & SYSTEMS PROJECTS

Jun 2026 – Present — **AgriGuard-RAG — End-to-End Agentic RAG System** — Live Demo | GitHub

- Engineered an end-to-end agentic RAG system over 5,277 official regulatory documents (E-Phy), deployed on Streamlit Cloud.
- **Semantic Intent Router:** Designed an upstream lightweight classifier (heuristics + TF-IDF / Logistic Regression) to dynamically categorize query complexity and eliminate redundant LLM calls.
- **Hybrid Search & RRF:** Combined dense representations (*FastEmbed*) and sparse retrieval (*BM25*) on **Qdrant Cloud**, reranked via **Reciprocal Rank Fusion (RRF)** with metadata pre-filtering.
- **Agentic Workflow:** Built automated multi-step query decomposition, parallel retrieval, context deduplication, and hallucination mitigation answer synthesis via the Groq Cloud API.
- **Technologies:** Python, Qdrant Cloud, Groq API, FastEmbed, BM25, Scikit-learn, Streamlit.

Nov 2025 — **Climate Sentiment Analyzer — Comparative Transformer Benchmark** — GitHub

- Evaluated sequence modeling architectures (RNN, LSTM vs. **BERT**) on noisy social media text data.
- Measured performance trade-offs across data regimes: analyzed attention gains in F1-score against inference latency and compute overhead relative to recurrent baselines.
- **Technologies:** PyTorch, Hugging Face Transformers, BERT, LSTM, Scikit-learn, Pandas.

Systems Project — **Full Modular Compiler (Object-Oriented Java Subset / DECA)** — GitHub

- Developed a modular compiler: lexical/syntactic parsing (**ANTLR**), AST construction, semantic verification, and assembly code generation.
- Focused on memory management, clean software architecture, and test-driven development with test suites (JaCoCo).
- **Technologies:** Java, ANTLR, Assembly, Register Allocation, Maven.

TECHNICAL SKILLS & LANGUAGES

LLM, Agents & RAG: Agentic Workflows, Hybrid Search (Dense + BM25), Qdrant Cloud, Reciprocal Rank Fusion (RRF), PEFT / LoRA, Representation Learning(JEPA), FastEmbed, LLM-as-a-Judge, Prompt Engineering, Hallucination Mitigation, LangChain, Reasoning Schemes (CoT, GoT), Router.
Machine Learning & NLP: PyTorch, Hugging Face Transformers, Scikit-learn, Dense/Sparse Embeddings, BERT, Tokenization, Feature Engineering, Dimensionality Reduction.
Software Engineering & Tools: Python, Java, C, C++, SQL, R, Linux, Git, Docker (basics), Streamlit Cloud, LaTeX.
Languages: Arabic: Native | **French:** Fluent | **English:** Professional(scientific literature, technical writing).